CNENGR202 Instrumentation and Control

Part 1 L8 – Feedback Control – Bode diagram

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Outline

- Fundamentals of Bode Diagrams
- Single pole response
- Single zero response
- Right half-plane zero
- Frequency inversion
- Inverted zero
- Combinations
- Example

Review Of Bode Plots - 1

- Decibels

$$|G|_{dB} = 20\log_{10}\left(|G|\right)$$

Expressing magnitudes in decibels

Actual magnitude	*Magnitude in dB*
1/10	-20dB
1/2	-6dB
1	0dB
2	6dB
10	20dB
100	40dB

Review OF Bode Plots - 2

- Decibels of quantities having units (impedance example): normalize before taking log

$$|Z|_{dB} = 20\log_{10}\left(\frac{|Z|}{R_{base}}\right)$$

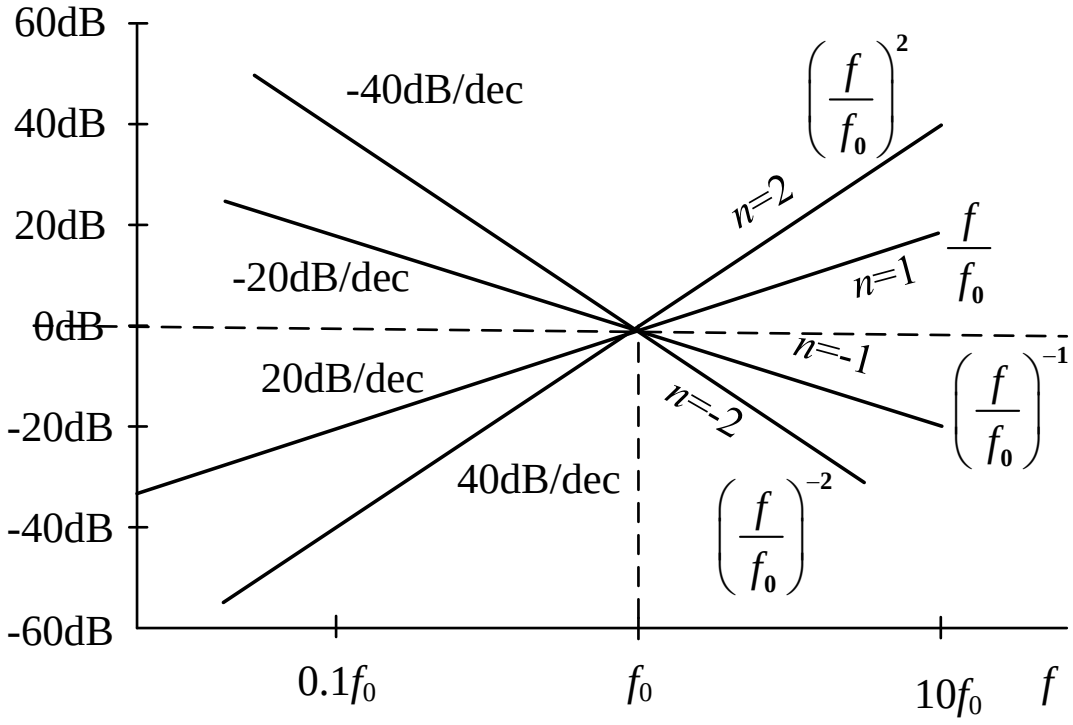
  - 2Ω is equivalent to 6dB with respect to a base impedance of R_{base} =1Ω, also known as 6dBΩ.

  - 60dBμA is a current 60dB greater than a base current of 1μA.

Review OF Bode Plots - 3

- Bode plots are effectively log-log plots, which cause functions which vary as f^n to become linear plots.

$$|G| = \left(\frac{f}{f_0}\right)^n$$

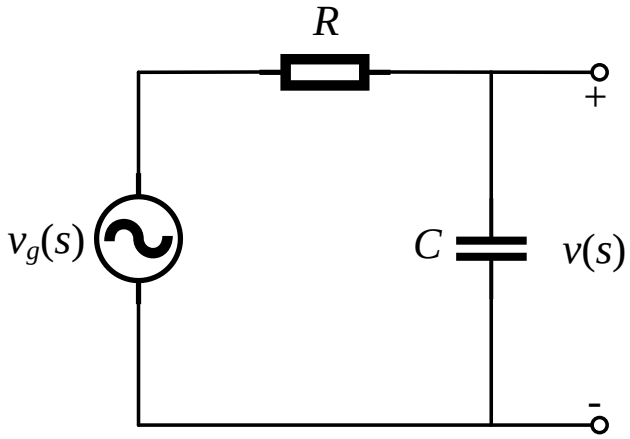


- Magnitude in dB is

$$|G|_{dB} = 20\log_{10}\left(\frac{f}{f_0}\right)^n = 20n\log_{10}\left(\frac{f}{f_0}\right)$$

- Slope is $20n$ dB/decade
- Magnitude is 1, or 0dB, at frequency $f = f_0$

Single pole response



- Simple *R*-*C* example
- Transfer function is

$$G(s)=\frac{v(s)}{v_g(s)}=\frac{\frac{1}{sC}}{R+\frac{1}{sC}}=\frac{1}{1+sRC}$$

- This coincides with the normalized form

$$G(s)=\frac{1}{1+\frac{s}{\omega_0}}$$

with $$\omega_0=\frac{1}{RC}$$

$G(j\omega)$ and $|G(j\omega)|$

- Let $s = j\omega$

$$G(j\omega) = \frac{1}{1 + j\frac{\omega}{\omega_0}} = \frac{1 - j\frac{\omega}{\omega_0}}{1 + \left(\frac{\omega}{\omega_0}\right)^2}$$

- Magnitude is

$$|G(j\omega)| = \sqrt{\left[\mathrm{Re}\left(G(j\omega)\right)\right]^2 + \left[\mathrm{Im}\left(G(j\omega)\right)\right]^2} = \frac{1}{\sqrt{1 + \left(\frac{\omega}{\omega_0}\right)^2}}$$

- Magnitude in dB

$$|G(j\omega)|_{dB} = -20\log_{10}\sqrt{1 + \left(\frac{\omega}{\omega_0}\right)^2} = -10\log_{10}\left(1 + \left(\frac{\omega}{\omega_0}\right)^2\right)$$

Asymptotic behavior: low frequency

$$\left|G(j\omega)\right| = \frac{1}{\sqrt{1+\left(\frac{\omega}{\omega_0}\right)^2}}$$

- For small frequency

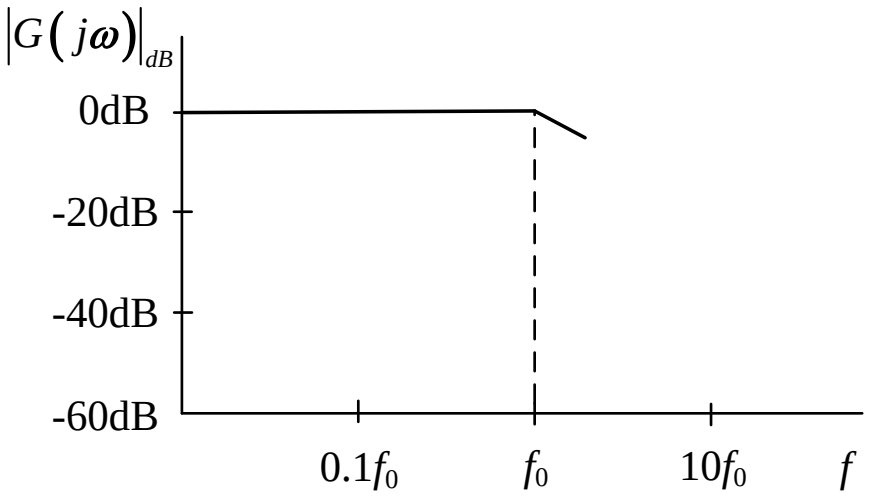
$$\omega \ll \omega_0$$

- Then $\left|G(j\omega)\right|$ becomes

$$\left|G(j\omega)\right| \approx \frac{1}{\sqrt{1}} = 1$$

- Or, in dB

$$\left|G(j\omega)\right|_{dB} \approx 0\text{dB}$$



This is the low-frequency asymptote of $\left|G(j\omega)\right|$

Asymptotic behavior: high frequency

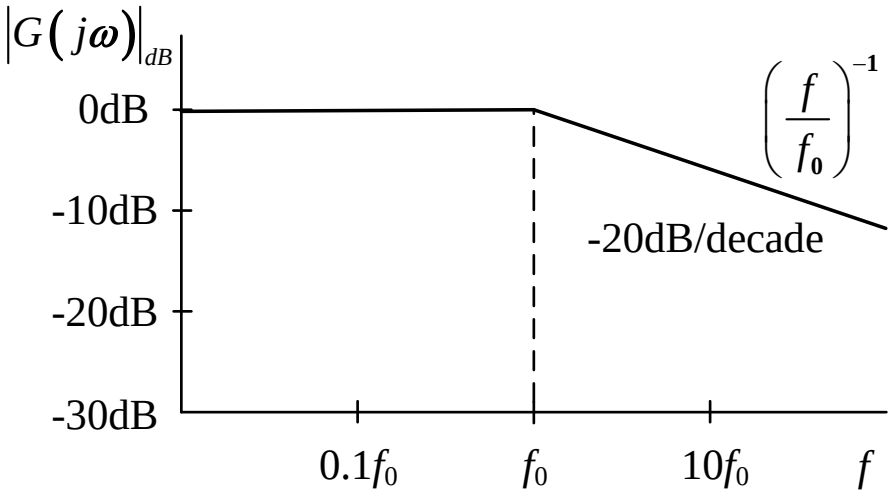
$$\left|G(j\omega)\right| = \frac{1}{\sqrt{1+\left(\frac{\omega}{\omega_0}\right)^2}}$$

- For high frequency

$$\omega \gg \omega_0$$

- Then $\left|G(j\omega)\right|$ becomes

$$\left|G(j\omega)\right| \approx \frac{1}{\sqrt{\left(\frac{\omega}{\omega_0}\right)^2}} = \left(\frac{f}{f_0}\right)^{-1}$$



- The high-frequency asymptote of $\left|G(j\omega)\right|$ varies as f^{-1}. Hence, n = -1, and a straight-line asymptote having a slope of -20dB/decade is obtained. The asymptote has a value of 0dB at $f = f_0$.

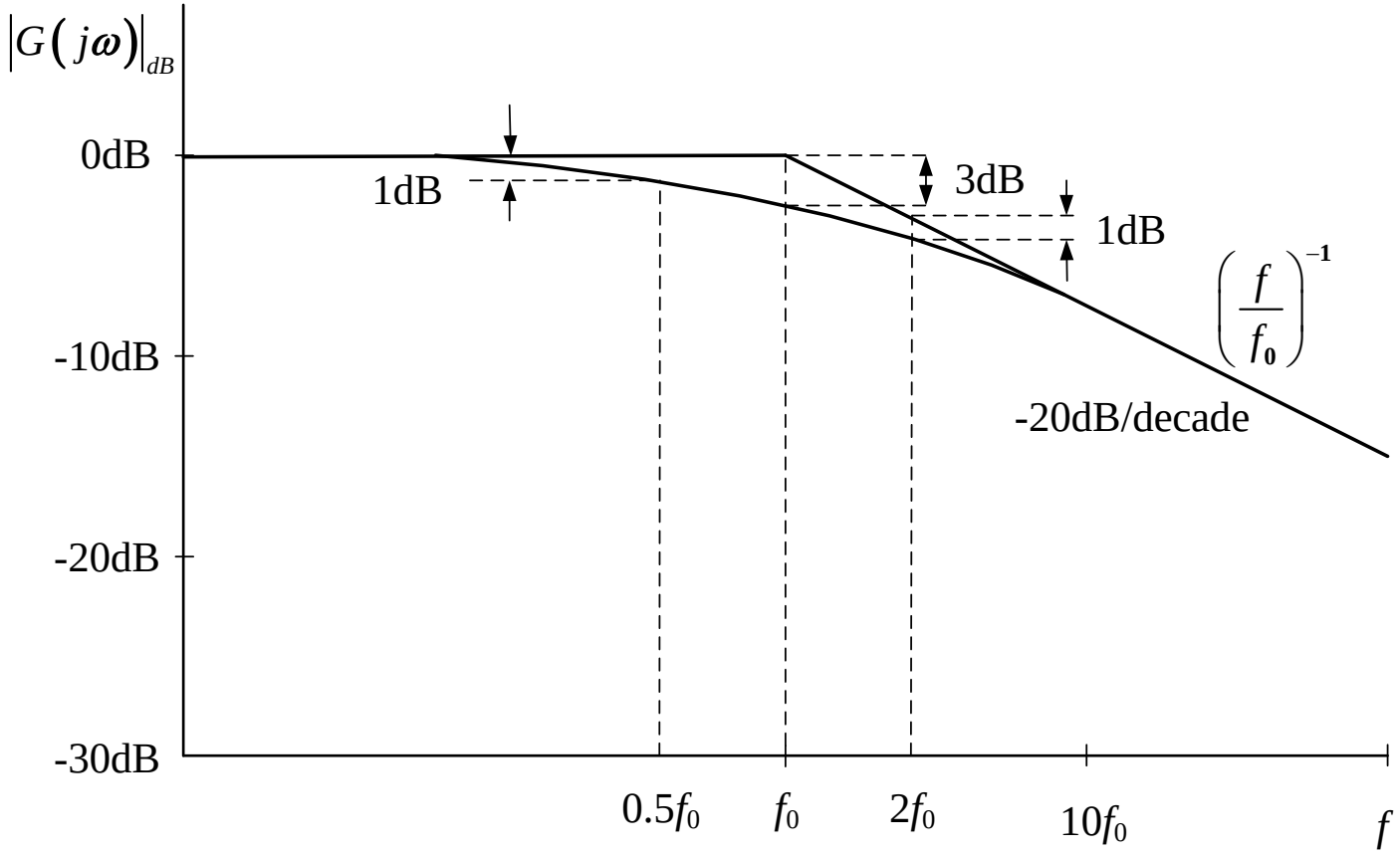
Deviation of exact curve near $f = f_0$

- Evaluate exact magnitude
  - at $f = f_0$

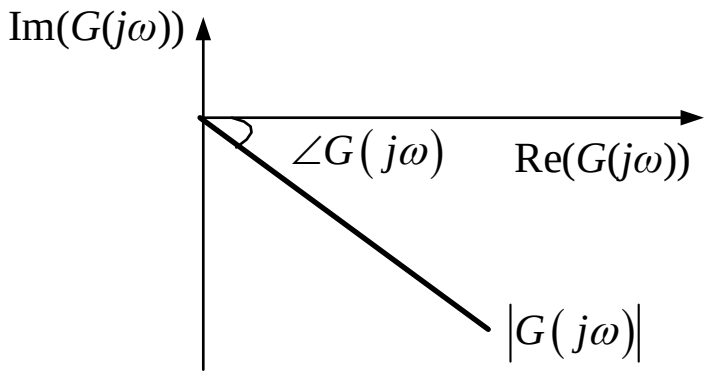
$$|G(j\omega)| = \frac{1}{\sqrt{1+\left(\frac{\omega_0}{\omega_0}\right)^2}} = \frac{1}{\sqrt{2}} \longrightarrow |G(j\omega)|_{dB} = -20\log_{10}\left(\sqrt{1+\left(\frac{\omega_0}{\omega_0}\right)^2}\right) \approx -3\text{dB}$$

  - at $f = \frac{1}{2} f_0$ and $2 f_0$: Similar arguments show that the exact curve lies 1dB below the asymptotes.

Summary: magnitude



Phase of $G(j\omega)$



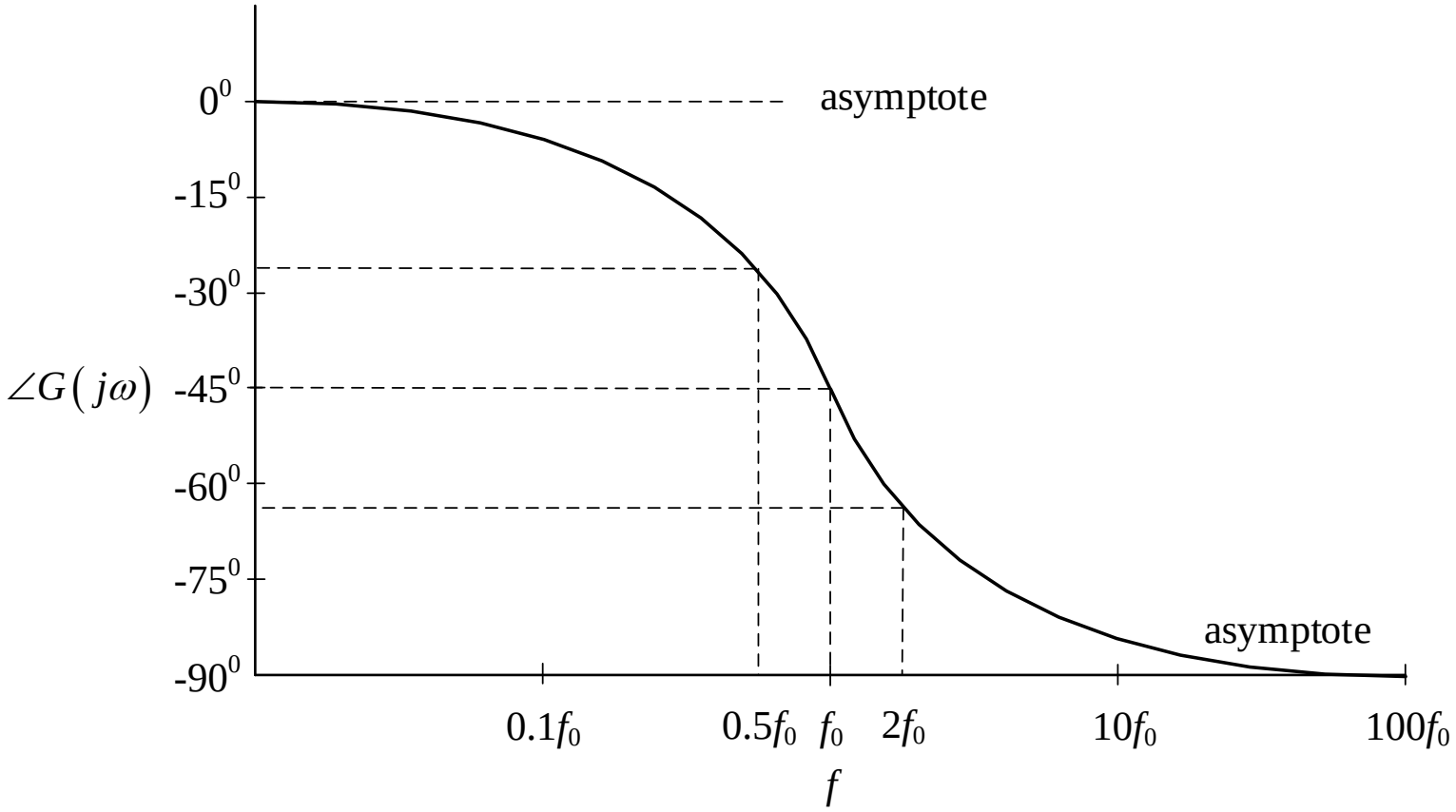
$$G(j\omega)=\frac{1}{1+j\dfrac{\omega}{\omega_0}}$$

$$\angle G(j\omega)=-\tan^{-1}\left(\frac{\omega}{\omega_0}\right)$$

$$\angle G(j\omega)=\tan^{-1}\left(\frac{\mathrm{Im}\left(G(j\omega)\right)}{\mathrm{Re}\left(G(j\omega)\right)}\right)$$

Phase of $G(j\omega)$

$$\angle G(j\omega) = -\tan^{-1}\left(\frac{\omega}{\omega_0}\right)$$



ω	$\angle G(j\omega)$
0	$\mathbf{0^0}$
$\boldsymbol{\omega_0}$	$\mathbf{-45^0}$
∞	$\mathbf{-90^0}$

Phase asymptotes

- Low frequency: $\mathbf{0^0}$
- High frequency: $\mathbf{-90^0}$
- Low- and high-frequency asymptotes do not intersect
  - Hence, need a midfrequency asymptote
- Try a midfrequency asymptote having slope identical to actual slope at the corner frequency f_0. One can show that the asymptotes then intersect at the break frequencies

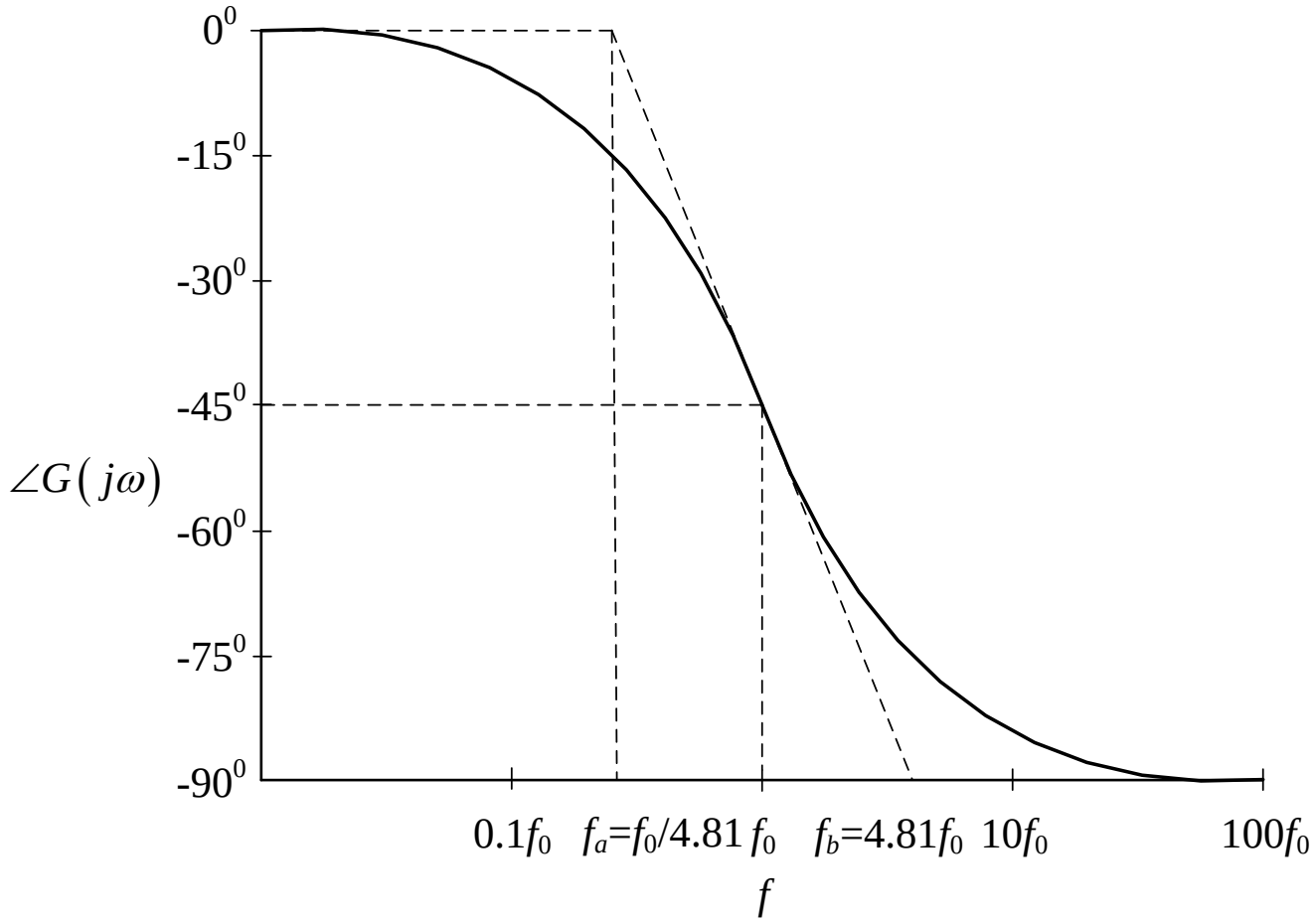
$$f_a = f_0 e^{-\pi/2} \approx f_0/4.81$$

$$f_b = f_0 e^{\pi/2} \approx 4.81\, f_0$$

Phase asymptotes

$$f_a = f_0 e^{-\pi/2} \approx f_0/4.81$$

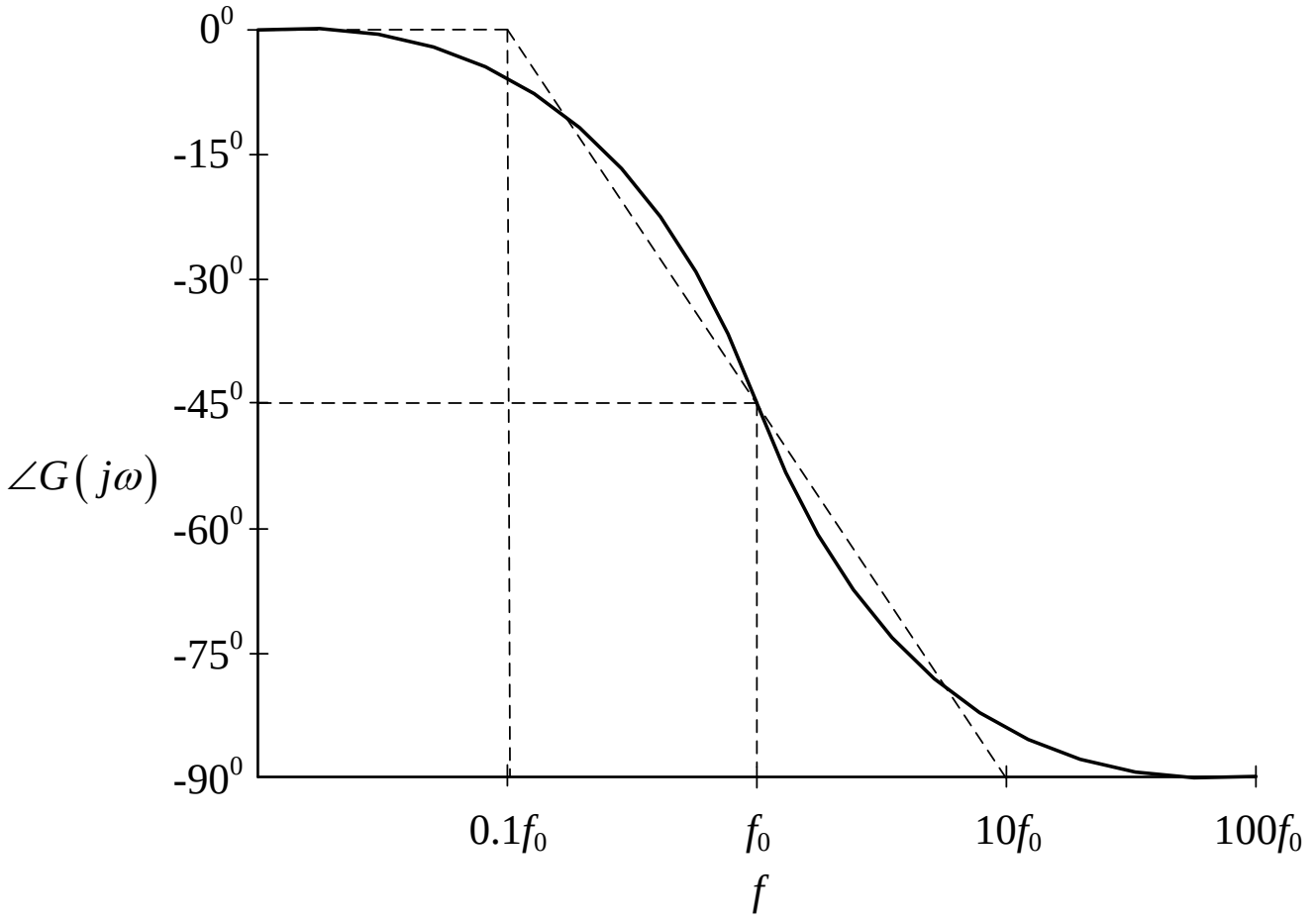
$$f_b = f_0 e^{\pi/2} \approx 4.81 f_0$$



Phase asymptotes: a simpler choice

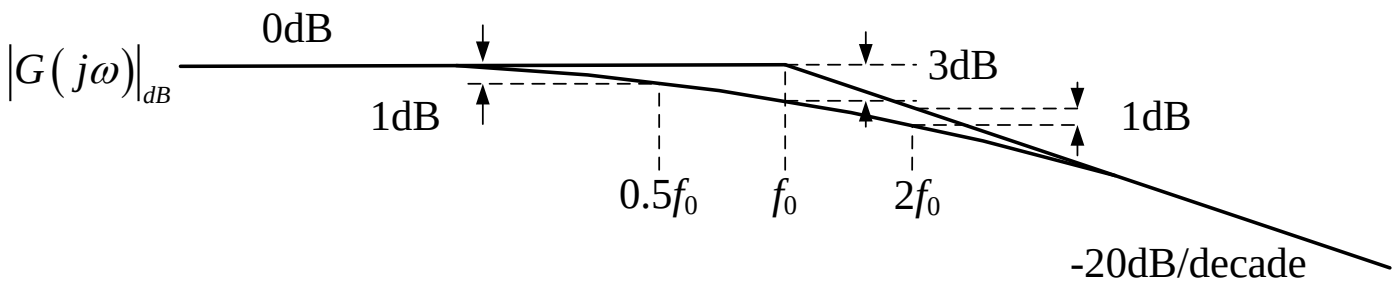
$$f_a = f_0/10$$

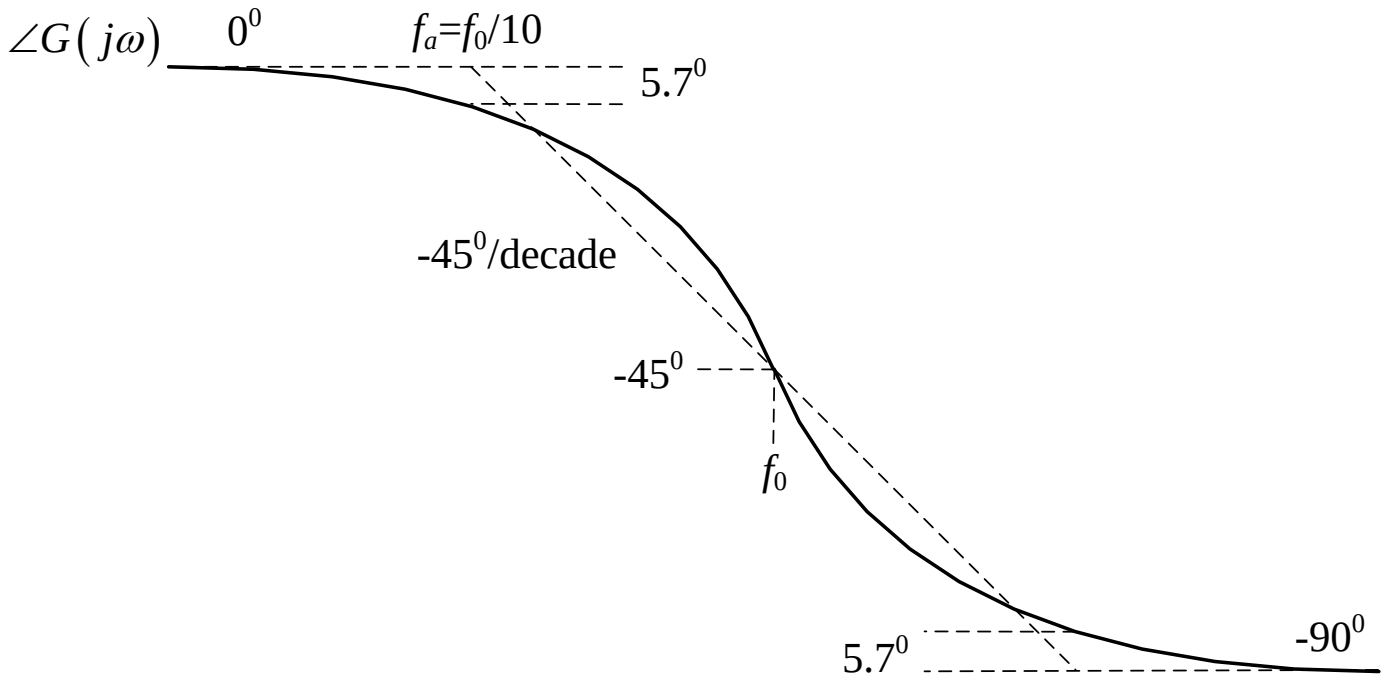
$$f_b = 10\, f_0$$



Summary: Bode plot of real pole

$$G(s)=\frac{1}{1+\frac{s}{\omega_0}}$$





Single zero response - 1

- Normalized form $G(s) = 1 + \frac{s}{\omega_0}$
- Magnitude $\left|G(j\omega)\right| = \sqrt{1 + \left(\frac{\omega}{\omega_0}\right)^2}$
- Use arguments similar to those used for the simple pole, to derive asymptotes
  - 0dB at low frequency $\omega \ll \omega_0$
  - 20dB/decade slope at high frequency $\omega \gg \omega_0$

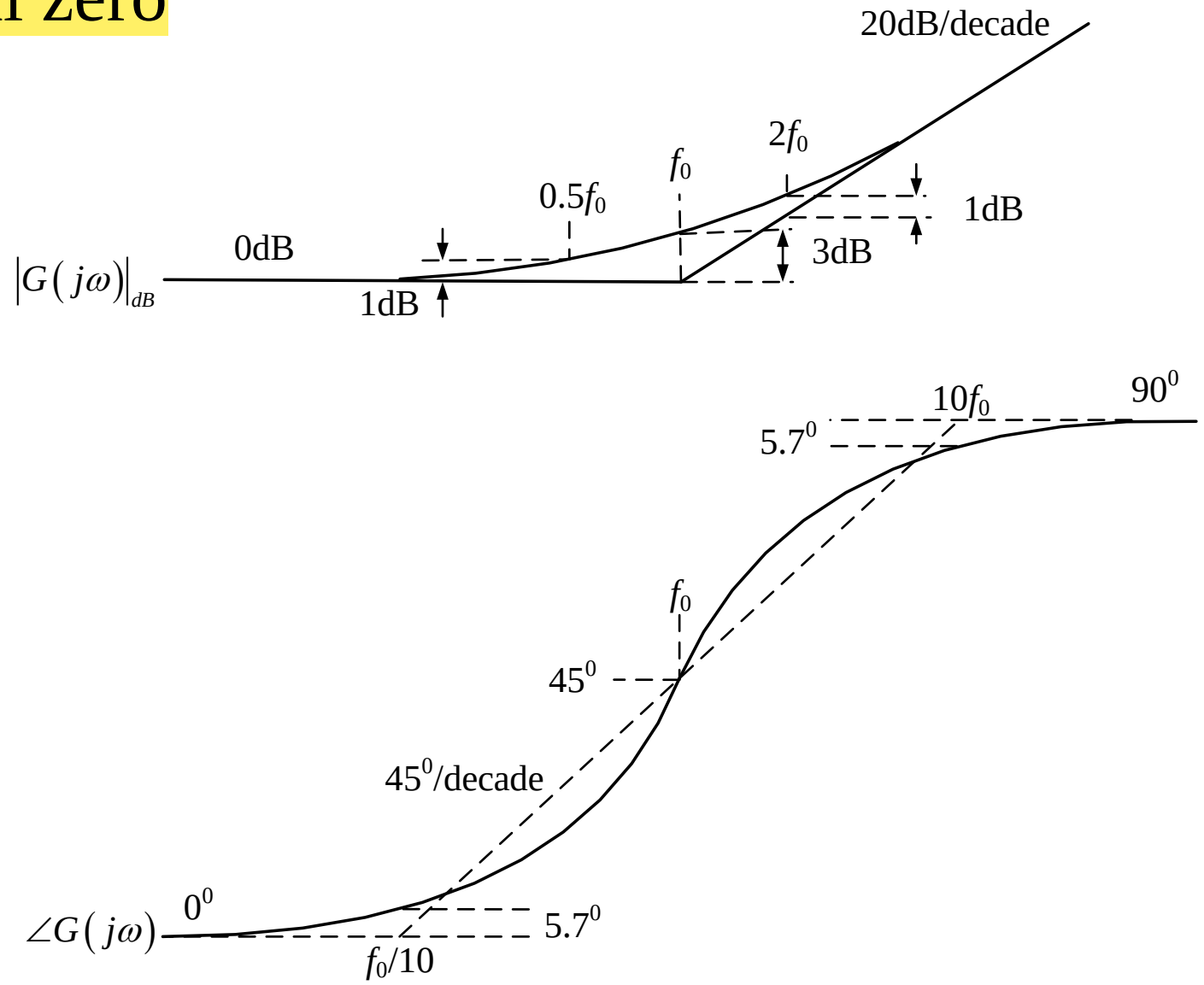
Single zero response - 2

- Phase

$$\angle G(j\omega) = \tan^{-1}\left(\frac{\omega}{\omega_0}\right)$$

  - with the exception of a missing minus sign, same as simple pole

Summary: Bode plot, real zero



Right half-plane zero

- Normalized form $G(s) = 1 - \dfrac{s}{\omega_0}$

- Magnitude $|G(j\omega)| = \sqrt{1 + \left(\dfrac{\omega}{\omega_0}\right)^2}$

  - same as conventional (left half-plane) zero. Hence, magnitude asymptotes are identical to those of LHP zero

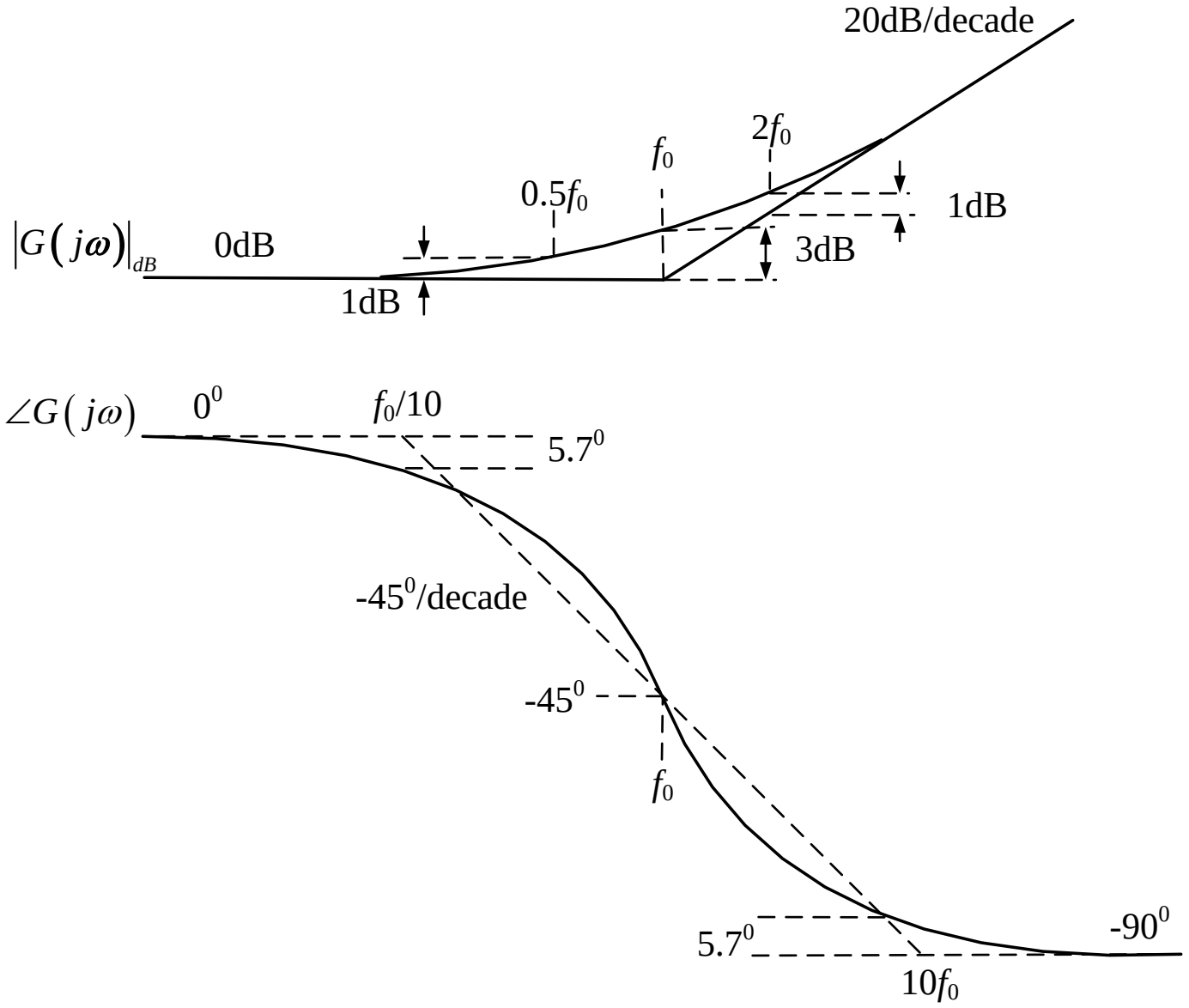
- Phase $\angle G(j\omega) = -\tan^{-1}\left(\dfrac{\omega}{\omega_0}\right)$

  - same as real pole

- The RHP zero exhibits the magnitude asymptotes of the LHP zero, and the phase asymptotes of the pole

Summary: Bode plot, RHP zero

$$G(s) = 1 - \frac{s}{\omega_0}$$



Frequency inversion

- Reversal of frequency axis. A useful form when describing mid- or high-frequency flat asymptotes. Normalized form, inverted pole

$$G(s)=\frac{1}{1+\frac{\omega_0}{s}}$$

- An algebraically equivalent form

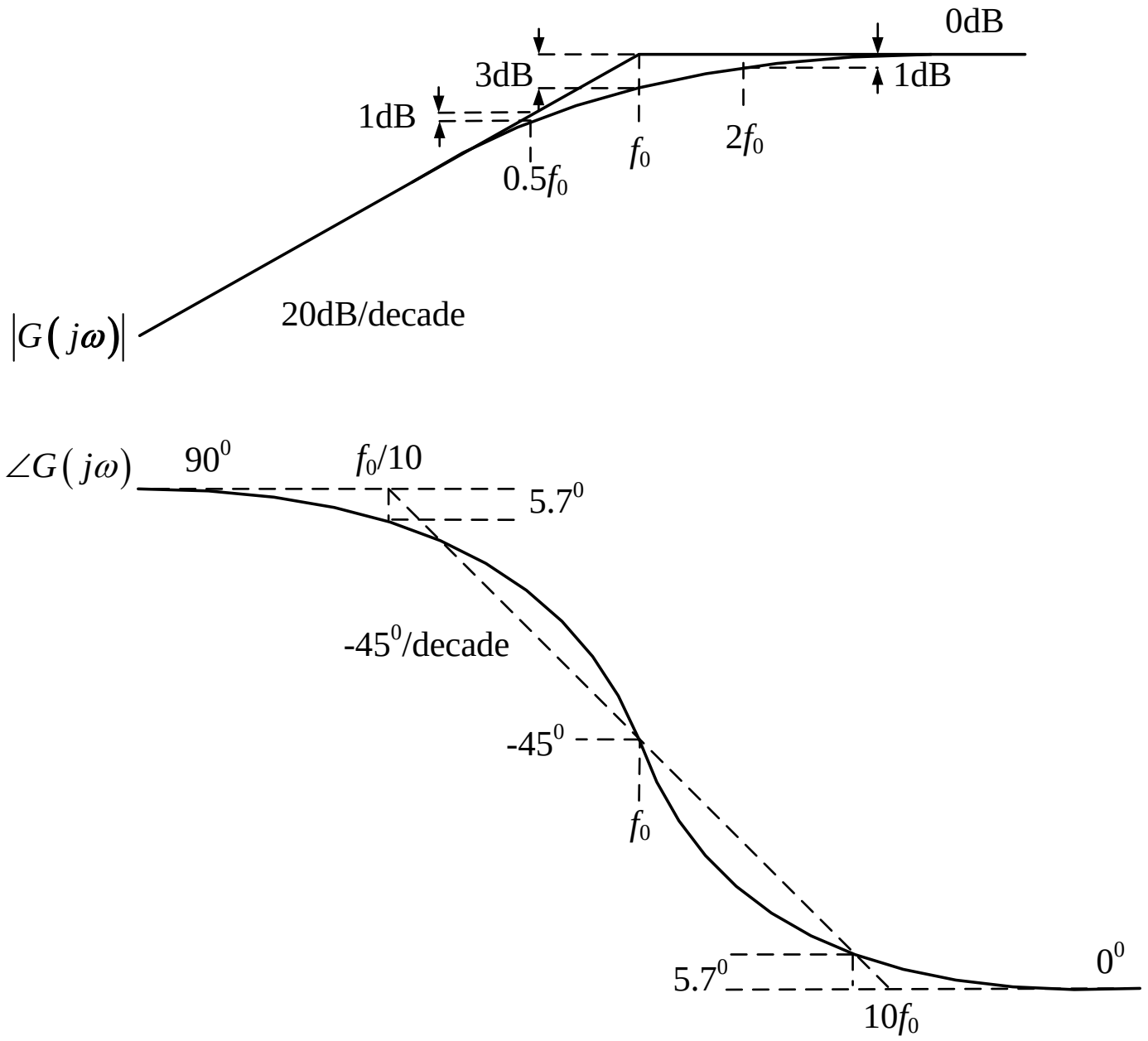
$$G(s)=\frac{\frac{s}{\omega_0}}{1+\frac{s}{\omega_0}}$$

- The inverted-pole format emphasizes the high-frequency gain

Asymptotes, inverted pole

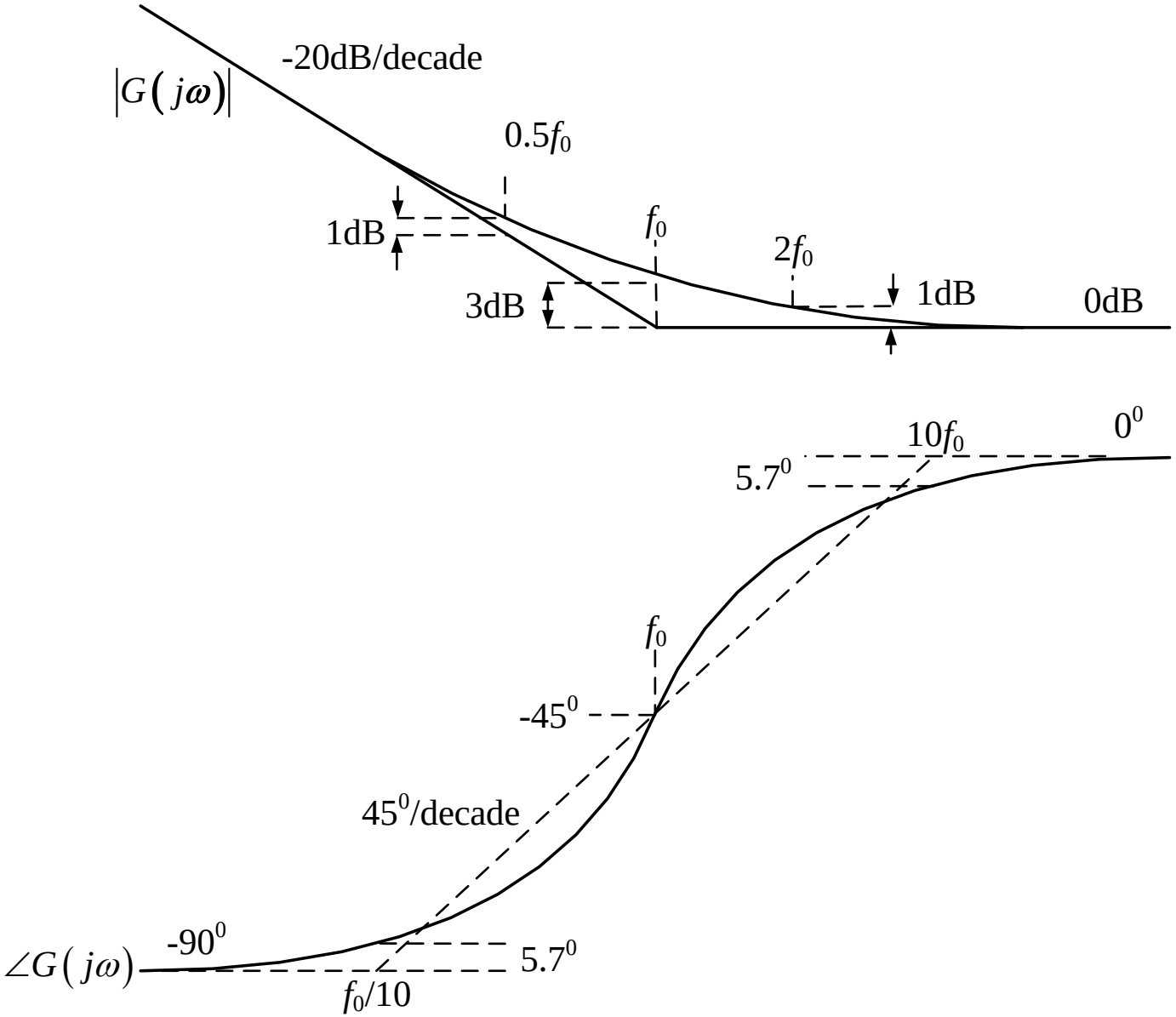
$$G(s)=\frac{1}{1+\frac{\omega_0}{s}}$$

$$=\frac{\frac{s}{\omega_0}}{1+\frac{s}{\omega_0}}$$



Inverted zero

$$G(s)=1+\frac{\omega_0}{s}$$



Combinations - 1

- Suppose that we have constructed the Bode diagrams of two complex-values functions of frequency, $G_1(\omega)$ and $G_2(\omega)$. It is desired to construct the Bode diagram of the product, $G_3(\omega) = G_1(\omega)\, G_2(\omega)$

- Express the complex-valued functions in polar form

$$G_1(j\omega) = R_1(\omega) e^{j\theta_1(\omega)}$$
$$G_2(j\omega) = R_2(\omega) e^{j\theta_2(\omega)}$$
$$G_3(j\omega) = R_3(\omega) e^{j\theta_3(\omega)}$$

- The product $G_3(\omega)$ can then be written

$$G_3(j\omega) = G_1(j\omega) G_2(j\omega) = R_1(\omega) e^{j\theta_1(\omega)} R_2(\omega) e^{j\theta_2(\omega)}$$
$$= R_1(\omega) R_2(\omega) e^{j(\theta_1(\omega)+\theta_2(\omega))}$$

Combinations - 2

- The composite phase is

$$\theta_3(\omega) = \theta_1(\omega) + \theta_2(\omega)$$

- The composite magnitude is

$$R_3(\omega) = R_1(\omega) R_2(\omega) \longrightarrow \left|R_3(\omega)\right|_{dB} = \left|R_1(\omega)\right|_{dB} + \left|R_2(\omega)\right|_{dB}$$

- Composite phase is sum of individual phases.
- Composite magnitude, when expressed in dB, is sum of individual magnitudes.

Example 1-1

$$G(s)=\frac{G_0}{\left(1+\frac{s}{\omega_1}\right)\left(1+\frac{s}{\omega_2}\right)}$$

- with

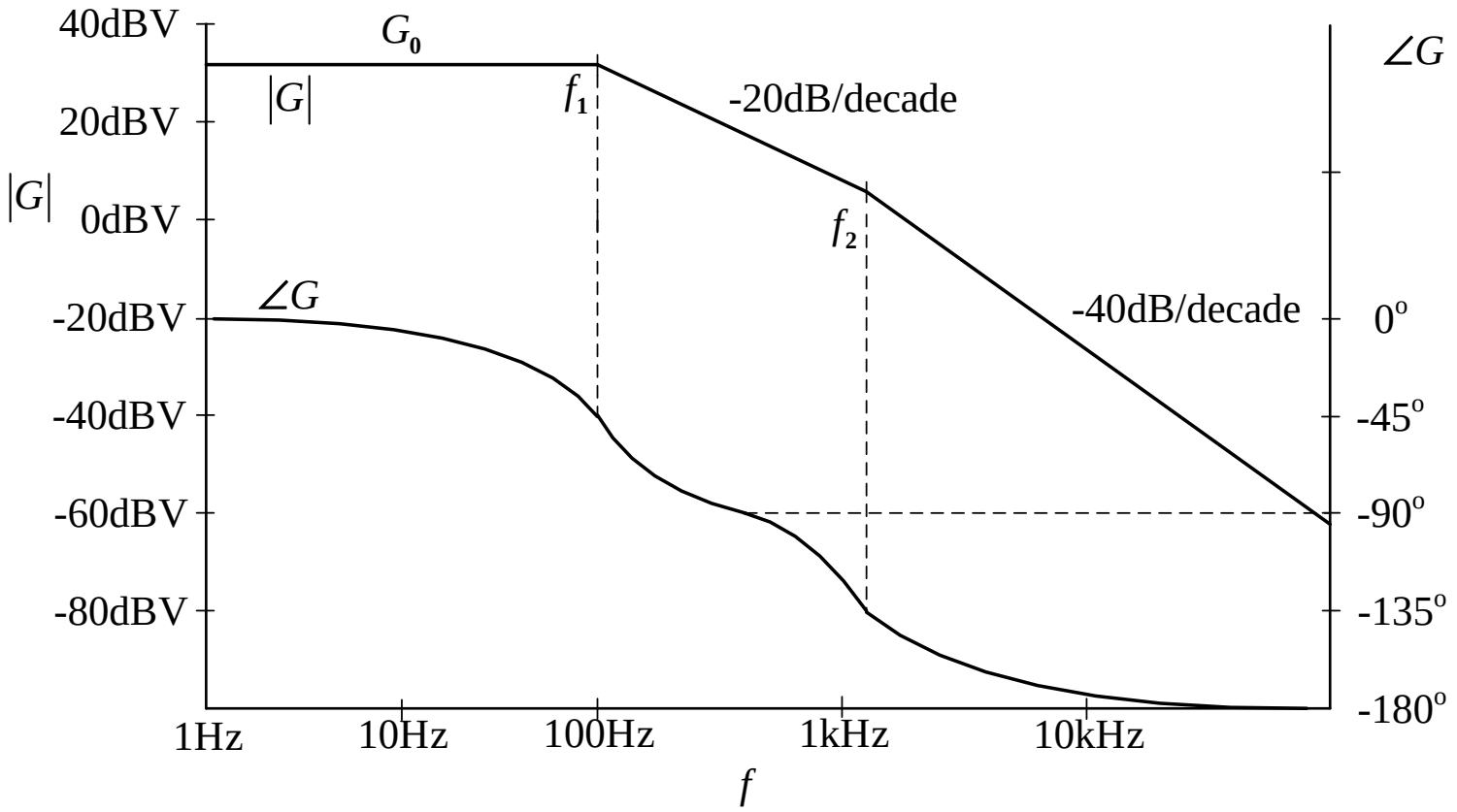
$$G_0=40\Rightarrow 32\text{dB}$$
$$f_1=\omega_1/2\pi=100\text{Hz}$$
$$f_2=\omega_2/2\pi=2\text{kHz}$$

Example 1-2

$$G(s)=\frac{G_0}{\left(1+\frac{s}{\omega_1}\right)\left(1+\frac{s}{\omega_2}\right)}$$

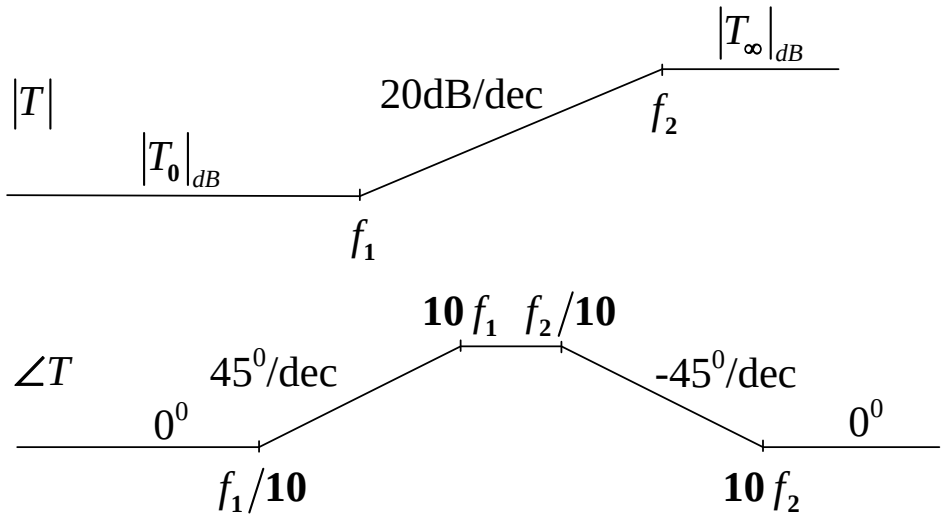
- with

$$G_0 = 40 \Rightarrow 32\text{dB}$$
$$f_1 = \omega_1/2\pi = 100\text{Hz}$$
$$f_2 = \omega_2/2\pi = 2\text{kHz}$$



Example 2 - 1

- Determine the transfer function $T(s)$ corresponding to the following asymptotes



Example 2 - 2

- Solution

$$T(s) = T_0 \frac{1+\frac{s}{\omega_1}}{1+\frac{s}{\omega_2}}$$

- Analytical expressions for asymptotes

  - For $f < f_1$

$$\left|T(s)\right|_{s=j\omega} = \left|T_0 \frac{1+\frac{s}{\omega_1}}{1+\frac{s}{\omega_2}}\right| = T_0$$

  - For $f_1 < f < f_2$

$$\left|T(s)\right|_{s=j\omega} = \left|T_0 \frac{1+\frac{s}{\omega_1}}{1+\frac{s}{\omega_2}}\right| = T_0 \frac{f}{f_1}$$

Example 2 - 3

- For $f > f_2$

$$\left|T(s)\right|_{s=j\omega} = \left| T_0 \frac{1+\dfrac{s}{\omega_1}}{1+\dfrac{s}{\omega_2}} \right| = T_0 \frac{f_2}{f_1}$$

- So the high-frequency asymptote is

$$T_\infty = T_0 \frac{f_2}{f_1}$$